

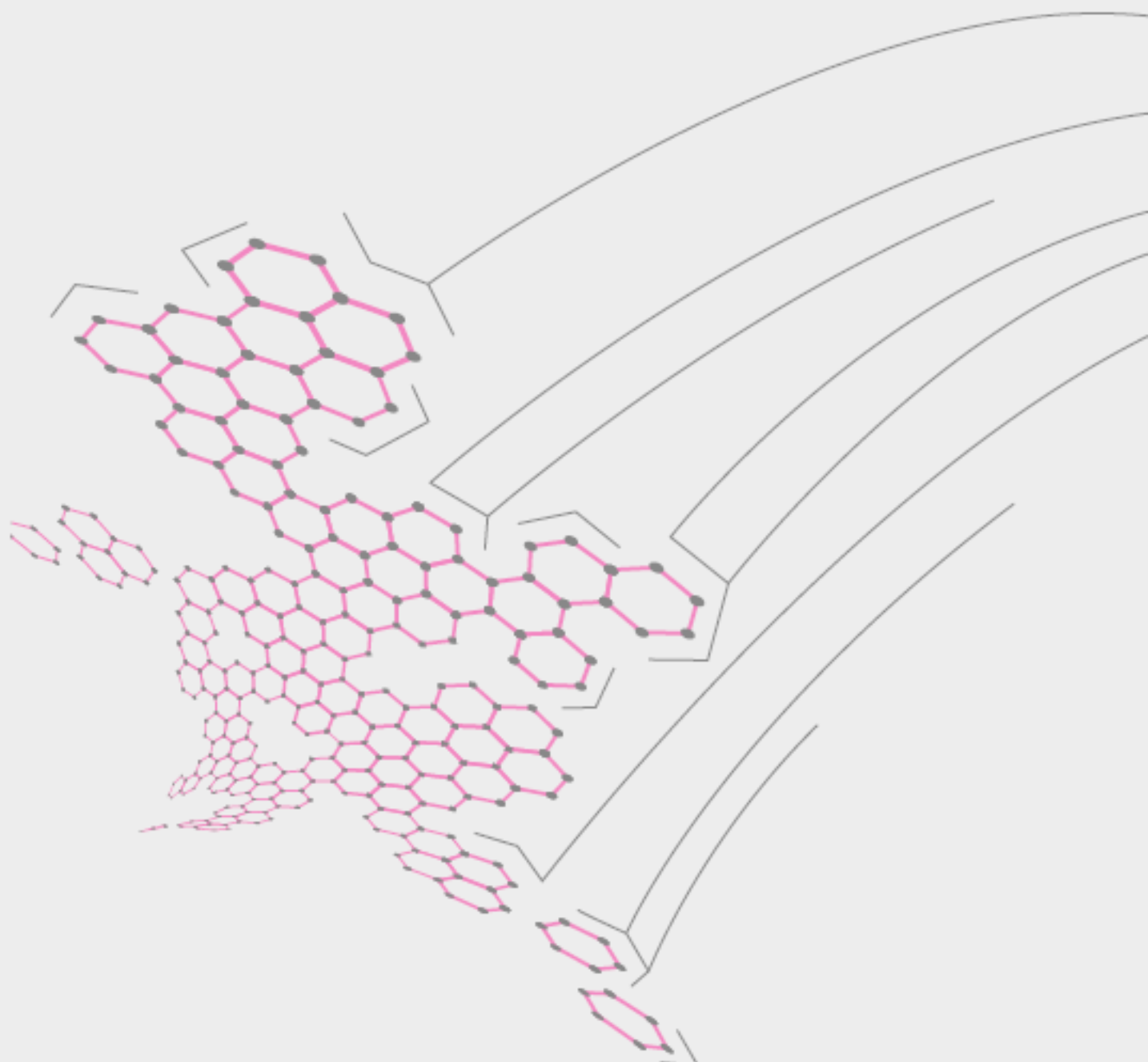
Calculus A: Lecture 1

Functions, limits, and continuity

Carlos Pérez Arancibia
(c.a.perezarancibia@utwente.nl)

Mathematics of Computational Science
Department of Applied Mathematics
University of Twente

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Functions: Basic idea

- One variable (y) depends on another (x).

- We write:

$$y = f(x) \quad (\text{"}y \text{ equals } f \text{ of } x\text{"})$$

- Notation:

- $f \rightarrow$ the function
- $x \rightarrow$ independent variable (input)
- $y \rightarrow$ dependent variable (output)



Example: Your final grade in this course:

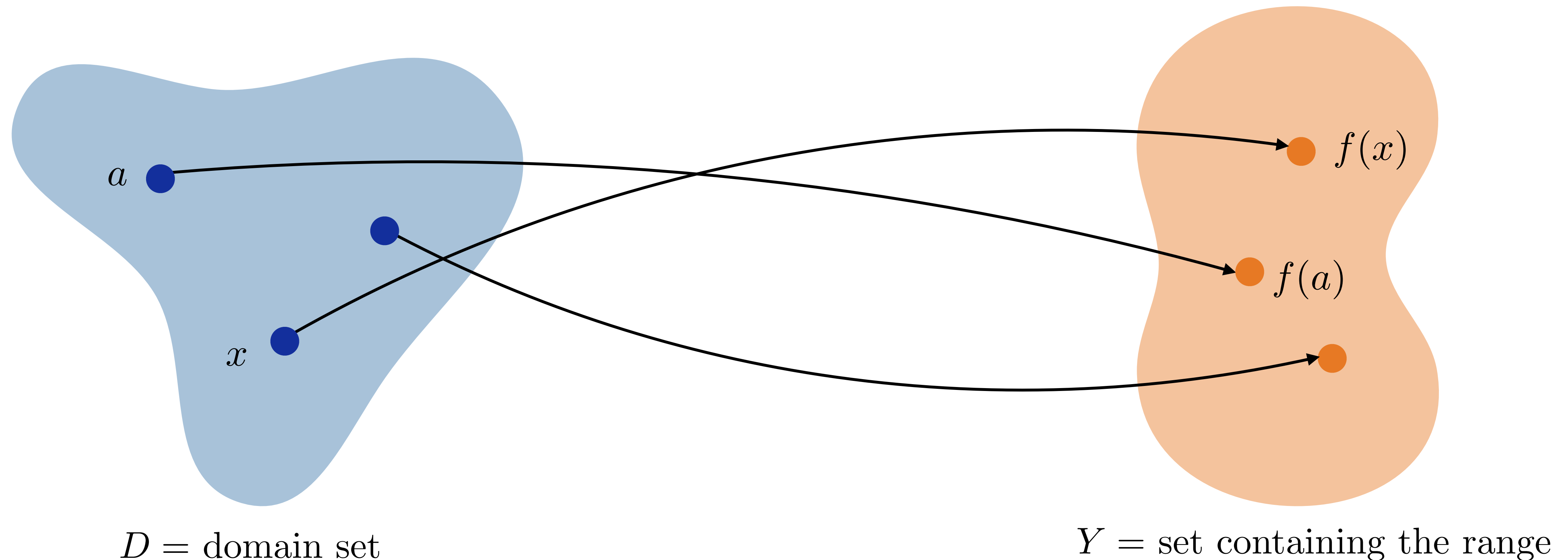
- Each student ID (x) is assigned exactly one final grade (y).
- The grading rule (e.g. $\text{grade} = \frac{\text{points earned by student } x}{\text{total points}} \times 9 + 1$) is the function f .
- Symbolically:

$y = f(x)$ means “grade y is the result of applying the rule to student x .”

Definition of a function

Definition:: A function f from a set D to a set Y is a rule that assigns a **unique** (single) element $f(x) \in Y$ to each element $x \in D$.

- The set D of all possible input values is the **domain**.
- The set of all outputs $f(x)$ for $x \in D$ is the **range** (not necessarily the entire target set Y).



Domain and range of a function

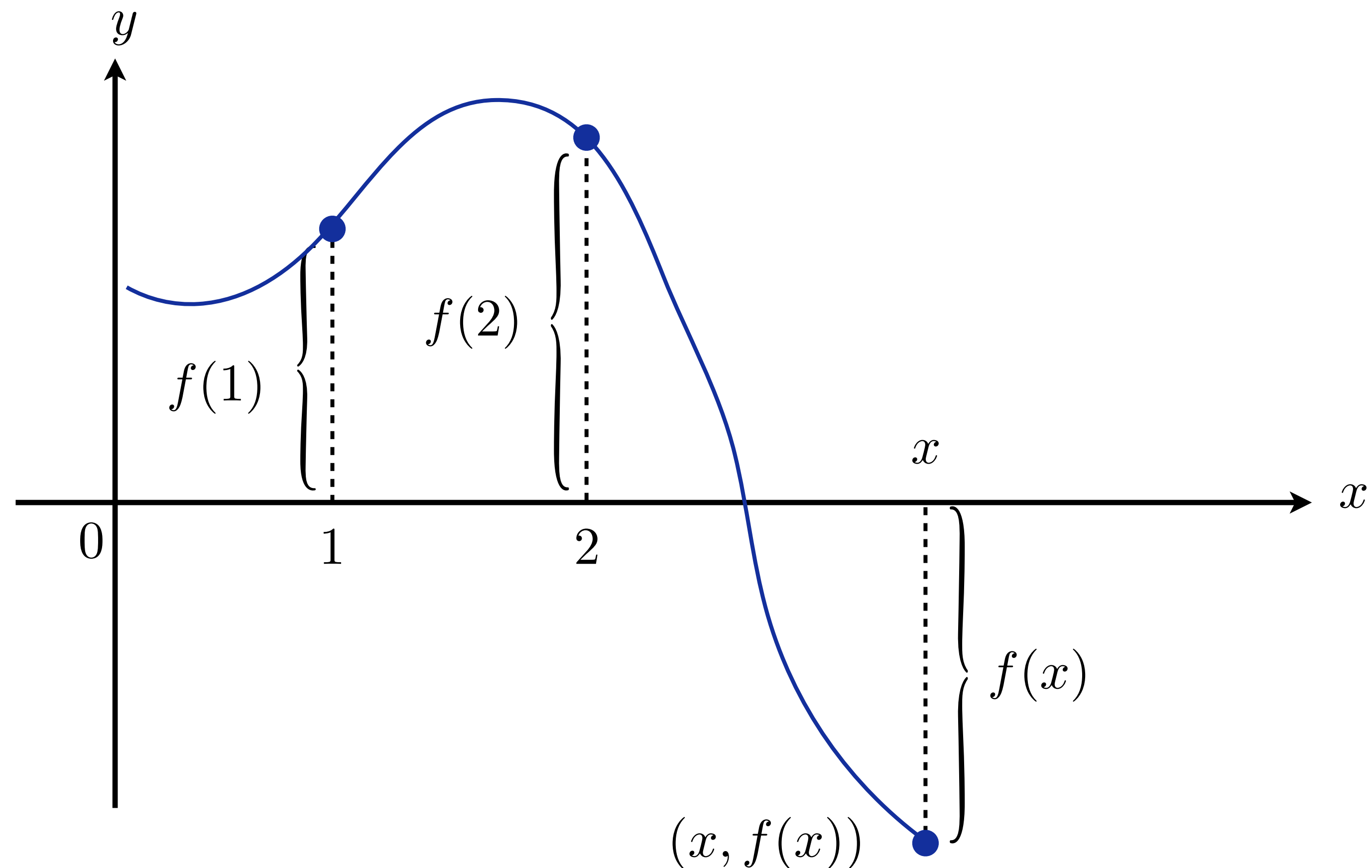
Example:

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$

Graph of a function

If f is a function with domain D , its **graph** is the set of points in the Cartesian plane corresponding to the input-output pairs of f . In set notation,

$$\{(x, f(x)) \mid x \in D\}.$$



Combining functions

Example:

Consider the functions

$$f(x) = \sqrt{x} \quad \text{and} \quad g(x) = \sqrt{1-x}$$

with respective domains:

$$D(f) = [0, \infty) \quad \text{and} \quad D(g) = (-\infty, 1]$$

Function	Formula	Domain
$f + g$	$(f + g)(x) = \sqrt{x} + \sqrt{1-x}$	$[0, 1] = D(f) \cap D(g)$
$f - g$	$(f - g)(x) = \sqrt{x} - \sqrt{1-x}$	$[0, 1]$
$g - f$	$(g - f)(x) = \sqrt{1-x} - \sqrt{x}$	$[0, 1]$
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1-x)}$	$[0, 1]$
f/g	$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1-x}}$	$[0, 1)$ ($x = 1$ excluded)
g/f	$\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1-x}{x}}$	$(0, 1]$ ($x = 0$ excluded)

Composition of functions

Definition: If f and g are functions, the composite function $f \circ g$ (“ f composed with g ”) is defined by

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ consists of the numbers x in the domain of g for which $g(x)$ lies in the domain of f .

Example: If $f(x) = \sqrt{x}$ and $g(x) = x + 1$, find

(a) $(f \circ g)(x)$ (b) $(g \circ f)(x)$ (c) $(f \circ f)(x)$ (d) $(g \circ g)(x)$.

Solution:

$$(a) \quad (f \circ g)(x) = f(g(x)) = f(x + 1) = \sqrt{x + 1}$$

$$(b) \quad (g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{x} + 1$$

$$(c) \quad (f \circ f)(x) = f(f(x)) = \sqrt{\sqrt{x}} = x^{1/4}$$

$$(d) \quad (g \circ g)(x) = g(g(x)) = x + 2$$

Limit of a function

Sometimes we want to understand how a function behaves **near** a particular point, even if the function is not defined exactly there.

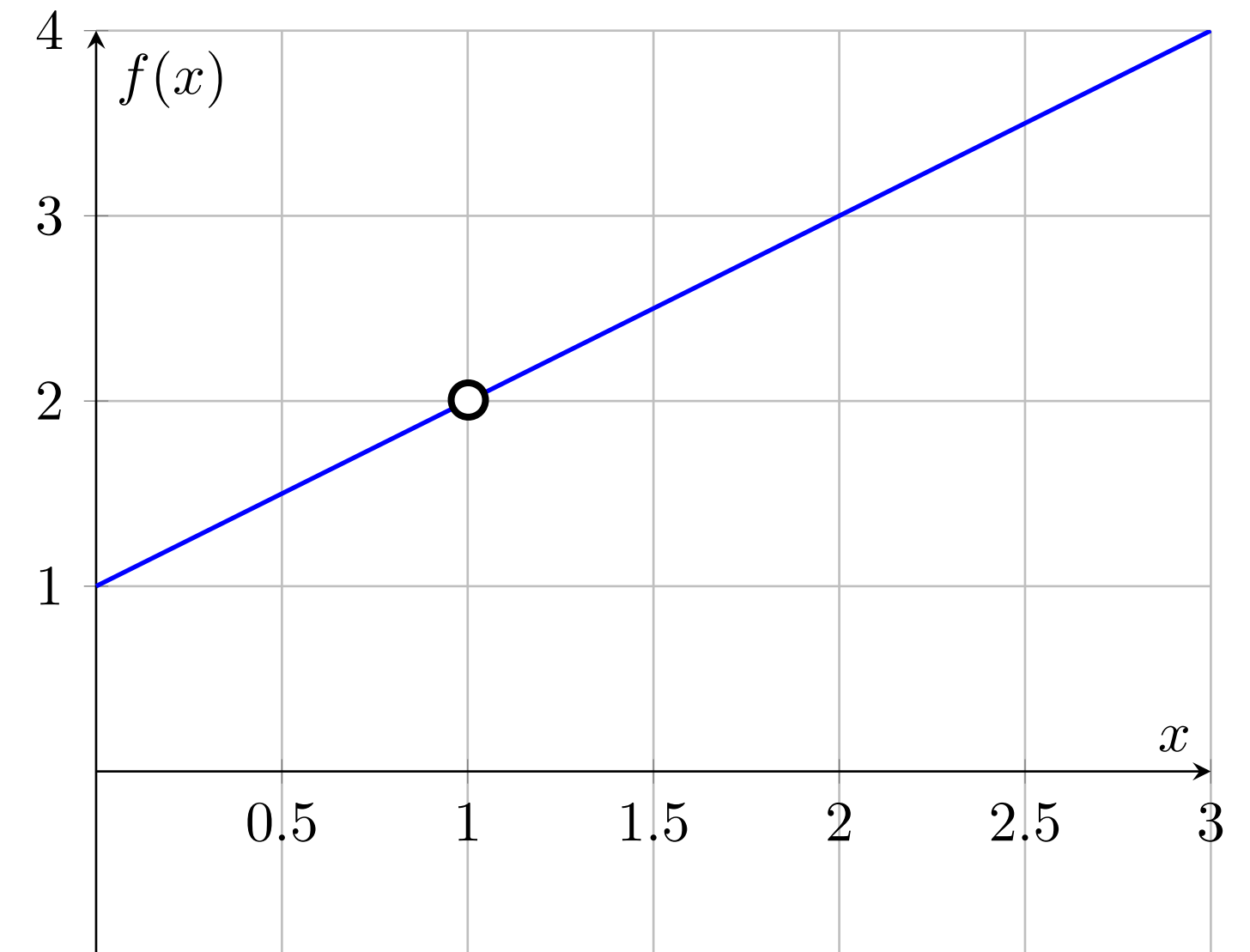
Example: Function with a “hole”.

Let

$$f(x) = \frac{x^2 - 1}{x - 1}$$

- The function is not defined at $x = 1$ because we would divide by zero.
- But for values close to 1, we can simplify:

$$f(x) = \frac{(x - 1)(x + 1)}{x - 1} = x + 1 \text{ for } x \neq 1$$



- So as x gets close to 1, $f(x)$ approaches 2, even though $f(1)$ **does not exist**.

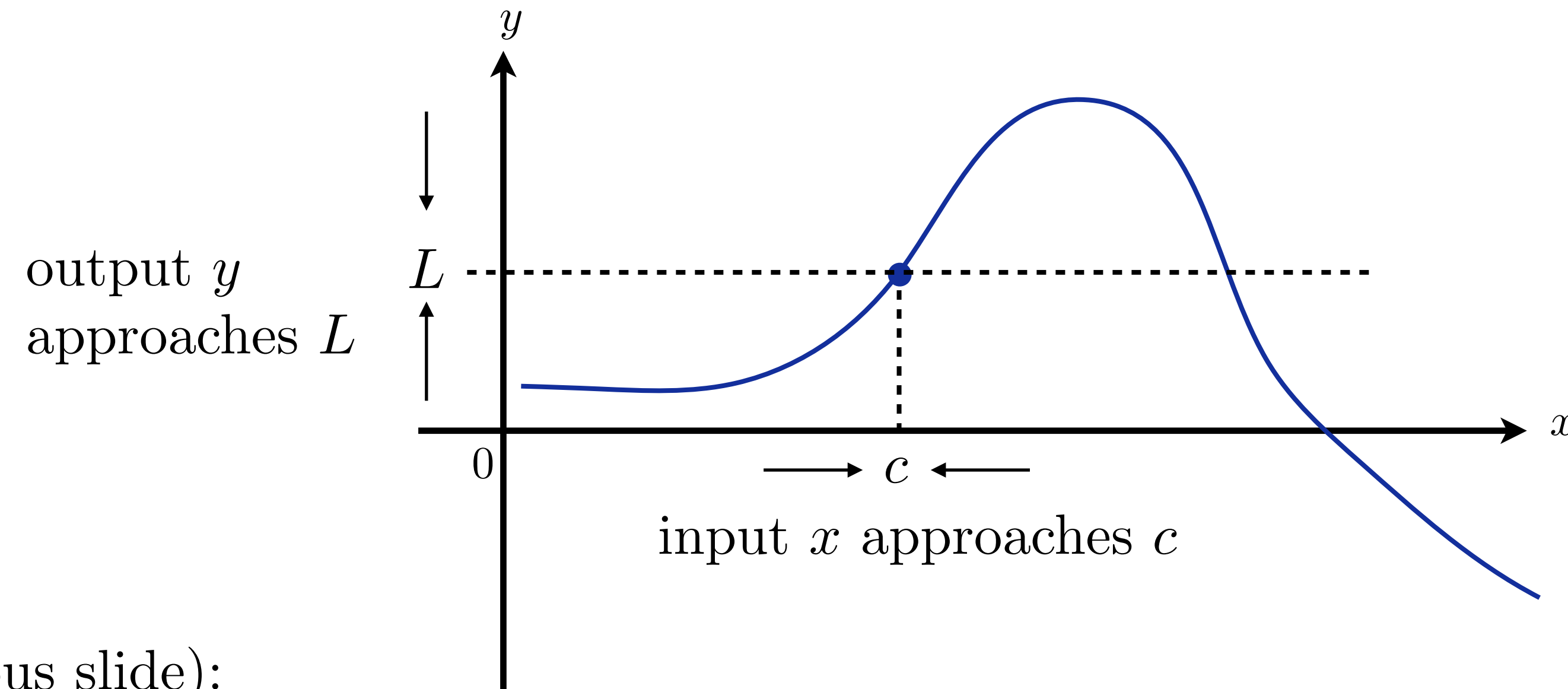
Observations like this naturally lead to the definition of **limit**.

Limit of a function

Notation:

$$\lim_{x \rightarrow c} f(x) = L$$

- Read as: “The limit of $f(x)$ as x approaches c is L .”
- Meaning: As x gets closer and closer to c , the values of $f(x)$ get closer and closer to L .



Example: (from previous slide):

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$$

Even though $f(1)$ is not defined, the limit exists because $f(x)$ approaches 2 as $x \rightarrow 1$.

Limit of a function

Theorem (Limit Laws): If L, M, c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then:

1. Sum Rule: $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$
2. Difference Rule: $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$
3. Constant Multiple Rule: $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$
4. Product Rule: $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$
5. Quotient Rule: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$
6. Power Rule: $\lim_{x \rightarrow c} [f(x)]^n = L^n, n \text{ a positive integer}$
7. Root Rule: $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ (If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

Limit of a function

Example: Evaluate the following limits:

$$\lim_{x \rightarrow -2} \sqrt{4x^2 - 3}, \quad \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}, \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$$

Solution:

$$\lim_{x \rightarrow -2} \sqrt{4x^2 - 3} = \sqrt{4(-2)^2 - 3} = \sqrt{16 - 3} = \sqrt{13}$$

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{(x-1)(x+2)}{x(x-1)} = \lim_{x \rightarrow 1} \frac{x+2}{x} = \frac{3}{1} = 3$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} &= \lim_{x \rightarrow 0} \frac{(\sqrt{x^2 + 100} - 10)(\sqrt{x^2 + 100} + 10)}{x^2(\sqrt{x^2 + 100} + 10)} \\ &= \lim_{x \rightarrow 0} \frac{x^2}{x^2(\sqrt{x^2 + 100} + 10)} = \frac{1}{20}. \end{aligned}$$

One-sided limits

Sometimes the value that a function approaches depends on the **direction** from which we approach a given point.

Example: Consider

$$f(x) = \frac{|x|}{x}, \quad x \neq 0.$$

For $x > 0$, $f(x) = 1$.

For $x < 0$, $f(x) = -1$.

At $x = 0$, the function is not defined.

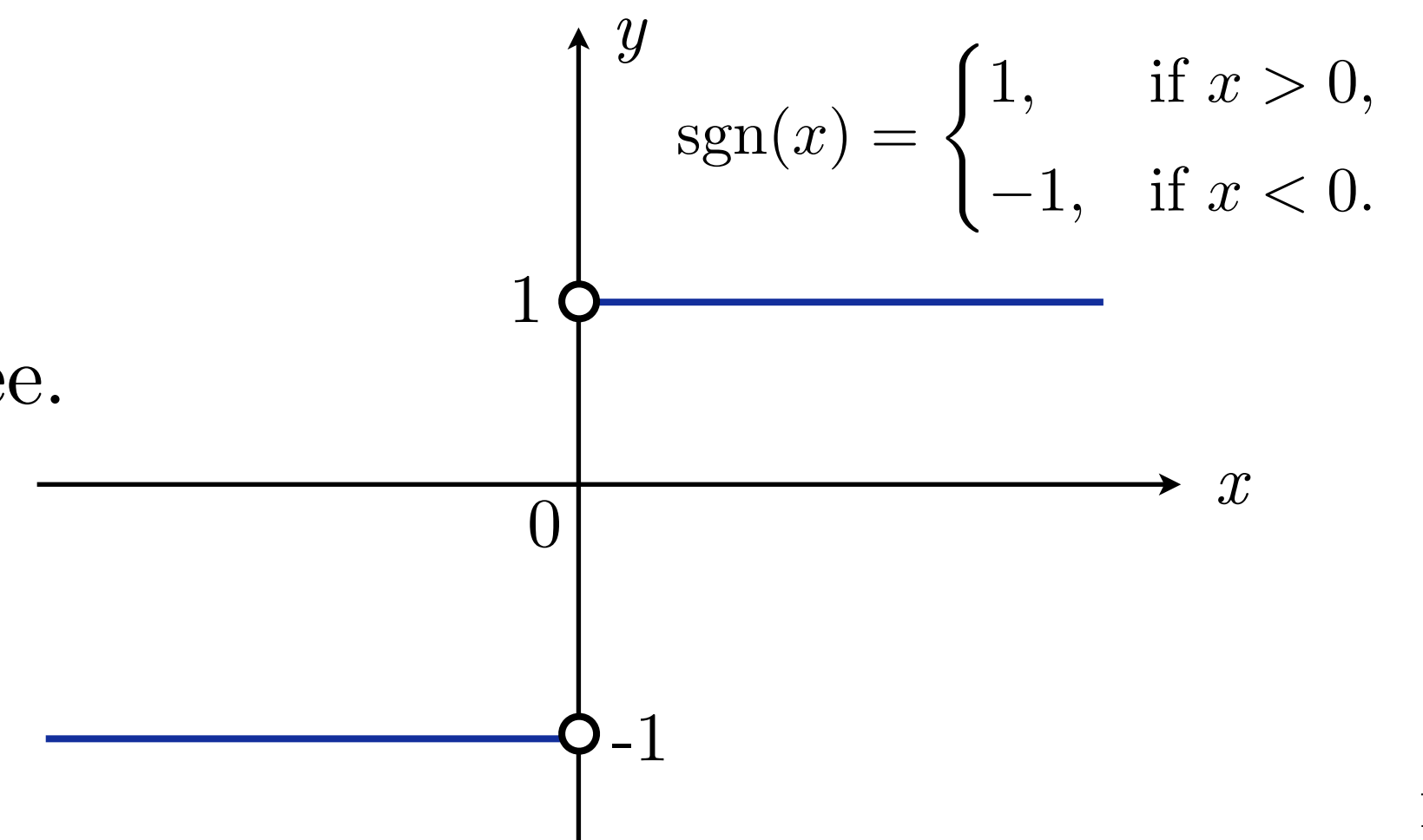
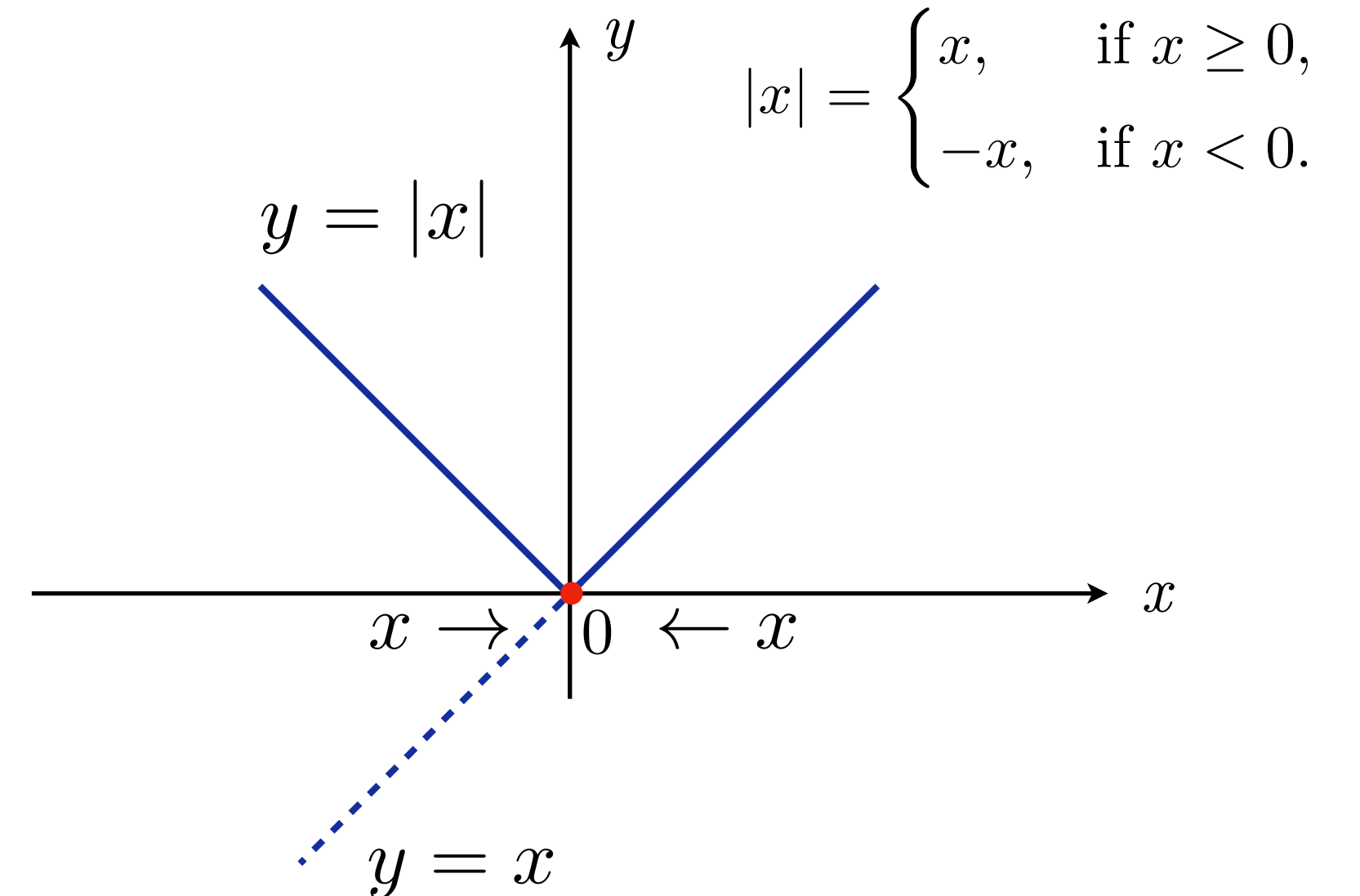
Therefore:

- If we approach 0 from the right ($x \rightarrow 0^+$), the values go to 1.
- If we approach 0 from the left ($x \rightarrow 0^-$), the values go to -1 .

Takeaway:

- The ordinary limit at 0 does not exist because the two sides disagree.
- To handle this properly, we define one-sided limits:

$$\lim_{x \rightarrow a^+} f(x) \quad \text{and} \quad \lim_{x \rightarrow a^-} f(x).$$

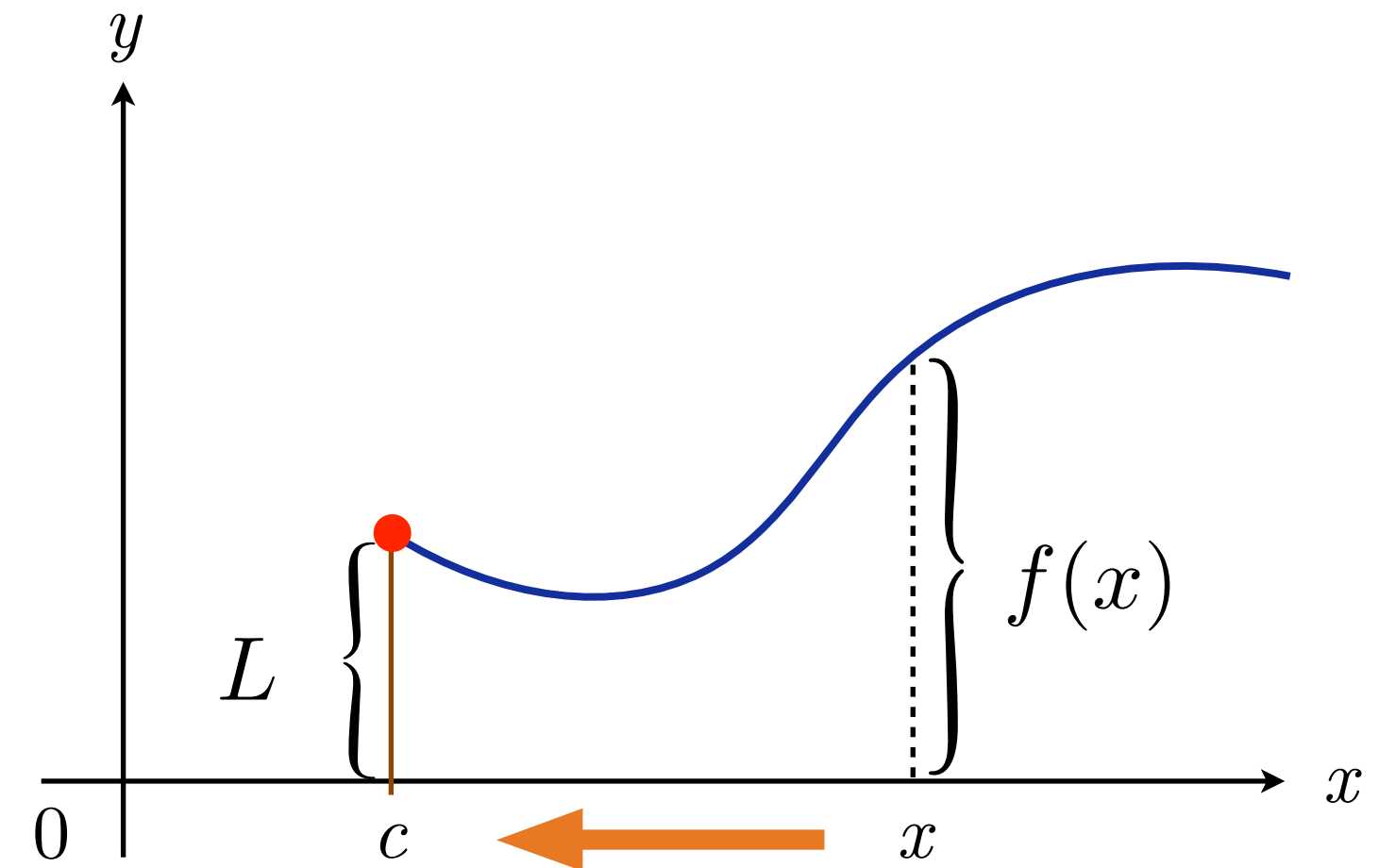


One-sided limits

If $f(x)$ is defined on (c, b) and $f(x)$ approaches L as x approaches c from the right, then we say that f has a right-hand limit L at c :

$$\lim_{x \rightarrow c^+} f(x) = L$$

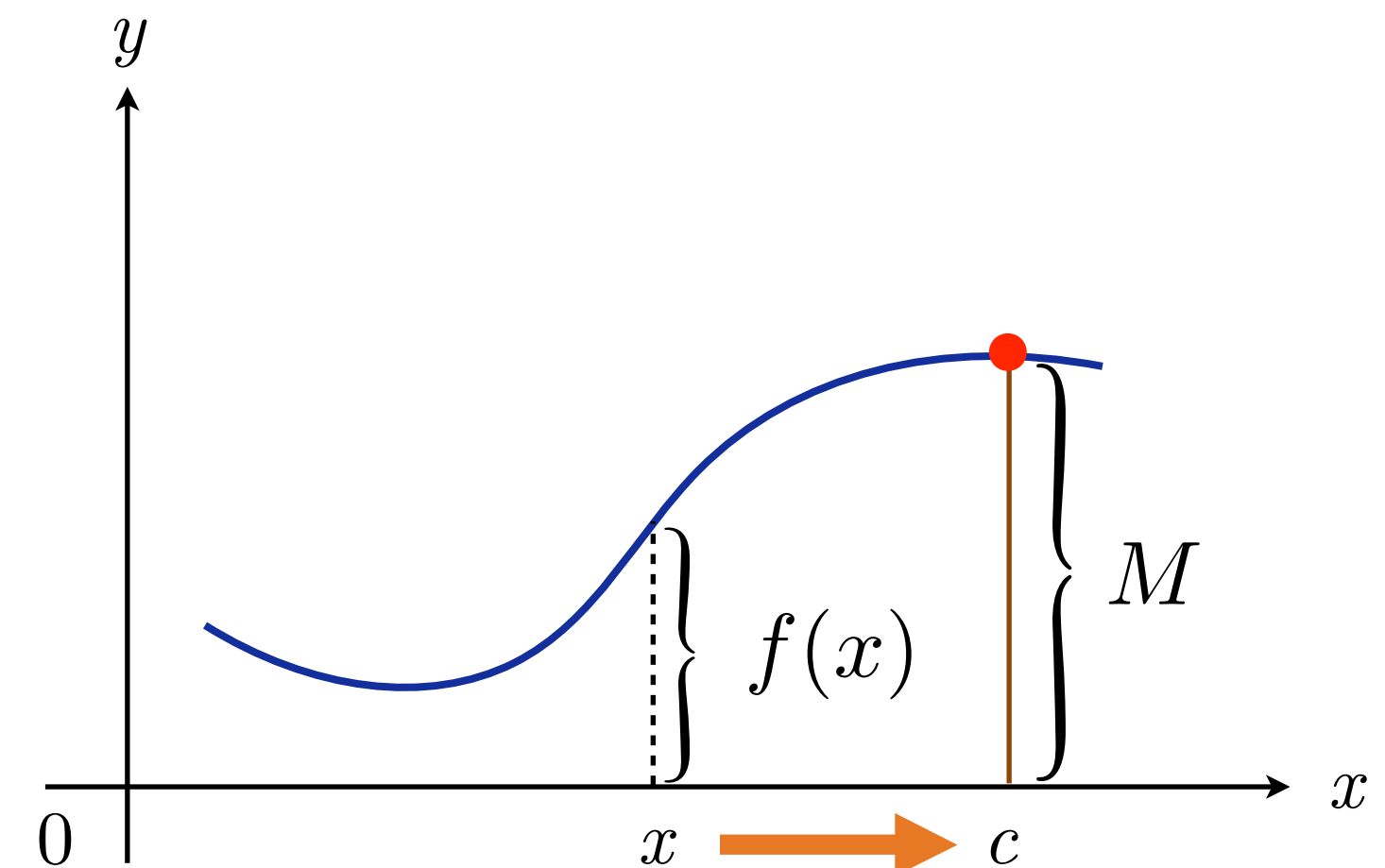
The notation $x \rightarrow c^+$ means x approaches c with $x > c$.



If $f(x)$ is defined on (a, c) and $f(x)$ approaches M as x approaches c from the left, then we say that f has a left-hand limit M at c :

$$\lim_{x \rightarrow c^-} f(x) = M$$

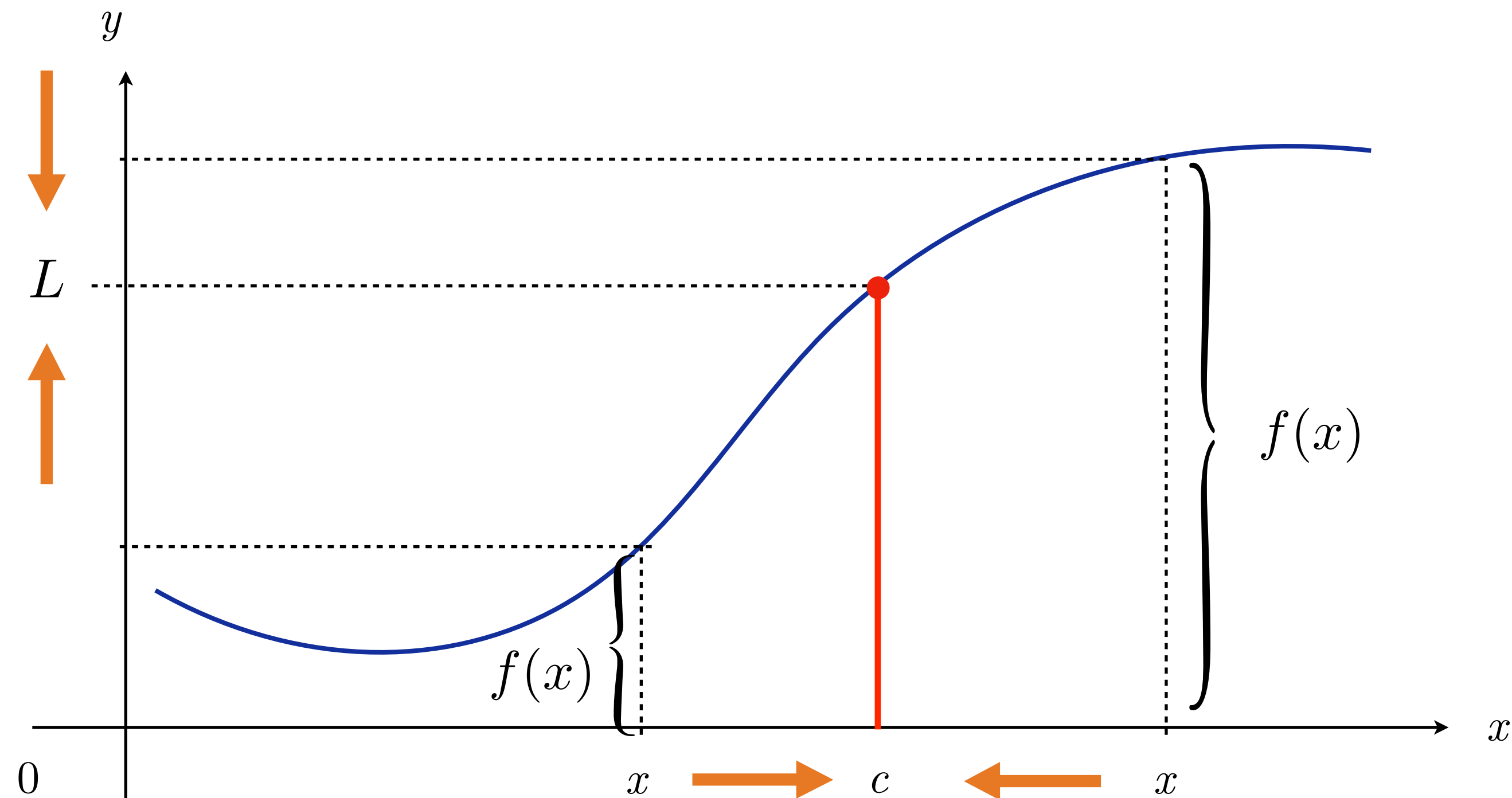
The notation $x \rightarrow c^-$ means x approaches c with $x < c$.



One-sided limits

Theorem: A function $f(x)$ has a limit as x approaches c if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x \rightarrow c} f(x) = L \quad \Longleftrightarrow \quad \left\{ \lim_{x \rightarrow c^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c^+} f(x) = L \right\}$$



Continuity

A function is continuous if, intuitively, its graph can be drawn without lifting the pencil.

To make this precise at a point c :

- The value of the function at c , $f(c)$, and
- The limit of $f(x)$ as $x \rightarrow c$,

must agree.

Formally, f is **continuous at c** if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

In other words, the function's value at c matches the value it is approaching as x gets close to c .

Continuity

Definition:

Interior point: A function $y = f(x)$ is continuous at an interior point c of its domain if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Endpoint: A function $y = f(x)$ is continuous at a left endpoint a or at a right endpoint b of its domain if

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{or} \quad \lim_{x \rightarrow b^-} f(x) = f(b),$$

respectively.

Continuity Test: A function $f(x)$ is continuous at an interior point $x = c$ of its domain if and only if it meets the following three conditions:

1. $f(c)$ exists (that is, c lies in the domain of f).
2. $\lim_{x \rightarrow c} f(x)$ exists (that is, f has a limit as $x \rightarrow c$).
3. $\lim_{x \rightarrow c} f(x) = f(c)$ (the limit equals the function value).

Continuity

Theorem (Properties of Continuous Functions):

If the functions f and g are continuous at $x = c$, then the following combinations are also continuous at $x = c$:

1. **Sums:** $f + g$
2. **Differences:** $f - g$
3. **Constant multiples:** $k \cdot f$, for any number k
4. **Products:** $f \cdot g$
5. **Quotients:** $\frac{f}{g}$, provided $g(c) \neq 0$
6. **Powers:** f^n , with n a positive integer
7. **Roots:** $\sqrt[n]{f}$, provided it is defined on an open interval containing c , with n a positive integer

Some common continuous functions

Constant and linear functions:

$f(x) = c$ (constant) $\rightarrow$ Continuous for all $x \in \mathbb{R}$

$f(x) = x$ (identity) $\rightarrow$ Continuous for all $x \in \mathbb{R}$

Polynomials and roots:

$f(x) = x^n, n \in \mathbb{N} \rightarrow$ Continuous for all $x \in \mathbb{R}$

$f(x) = \sqrt[n]{x} \rightarrow$ Continuous on domain

– even n : $x \geq 0$

– odd n : all $x \in \mathbb{R}$

Rational functions:

$f(x) = \frac{p(x)}{q(x)} \rightarrow$ Continuous wherever $q(x) \neq 0$

Exponentials and logarithms:

$f(x) = a^x, a > 0 \rightarrow$ Continuous for all $x \in \mathbb{R}$

$f(x) = e^x \rightarrow$ Continuous for all $x \in \mathbb{R}$

$f(x) = \ln(x) \rightarrow$ Continuous for $x > 0$

Trigonometric functions:

$\sin(x), \cos(x) \rightarrow$ Continuous for all $x \in \mathbb{R}$

$\tan(x) \rightarrow$ Continuous for $x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

$\cot(x) \rightarrow$ Continuous for $x \neq k\pi, k \in \mathbb{Z}$

$\sec(x) \rightarrow$ Continuous for $x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

$\csc(x) \rightarrow$ Continuous for $x \neq k\pi, k \in \mathbb{Z}$

Inverse trigonometric functions:

$\arcsin(x) \rightarrow$ Continuous for $x \in [-1, 1]$

$\arccos(x) \rightarrow$ Continuous for $x \in [-1, 1]$

$\arctan(x) \rightarrow$ Continuous for all $x \in \mathbb{R}$

Continuity

Theorem (Composite of Continuous Functions): If f is continuous at c and g is continuous at $f(c)$, then the composite $g \circ f$ is continuous at c .

Theorem (Limits of Continuous Functions): If g is continuous at the point b and $\lim_{x \rightarrow c} f(x) = b$, then

$$\lim_{x \rightarrow c} g(f(x)) = g(b) = g\left(\lim_{x \rightarrow c} f(x)\right).$$

Example: Show that the function

$$y = \sqrt{x^2 - 2x - 5}$$

is continuous on its domain.

Example: Compute

$$\lim_{x \rightarrow \pi/2} \cos\left(2x + \sin\left(\frac{3\pi}{2} + x\right)\right).$$

Continuity

Example: Show that the function

$$y = \sqrt{x^2 - 2x - 5}$$

is continuous on its domain.

Solution.

- The polynomial $p(x) = x^2 - 2x - 5$ is continuous on $\mathbb{R}$.
- The square root function $\sqrt{\cdot}$ is continuous on $[0, \infty)$.
- Therefore, $y = \sqrt{p(x)}$ is continuous wherever $p(x) \geq 0$.

Find the domain by solving

$$x^2 - 2x - 5 \geq 0 \quad \implies \quad x \leq 1 - \sqrt{6} \quad \text{or} \quad x \geq 1 + \sqrt{6}.$$

Hence, y is continuous on

$$(-\infty, 1 - \sqrt{6}] \cup [1 + \sqrt{6}, \infty).$$

Continuity

Example: Compute

$$\lim_{x \rightarrow \pi/2} \cos \left(2x + \sin \left(\frac{3\pi}{2} + x \right) \right).$$

Solution. Since $\cos$ and $\sin$ are continuous, we can move the limit inside:

$$= \cos \left(\lim_{x \rightarrow \pi/2} \left(2x + \sin \left(\frac{3\pi}{2} + x \right) \right) \right).$$

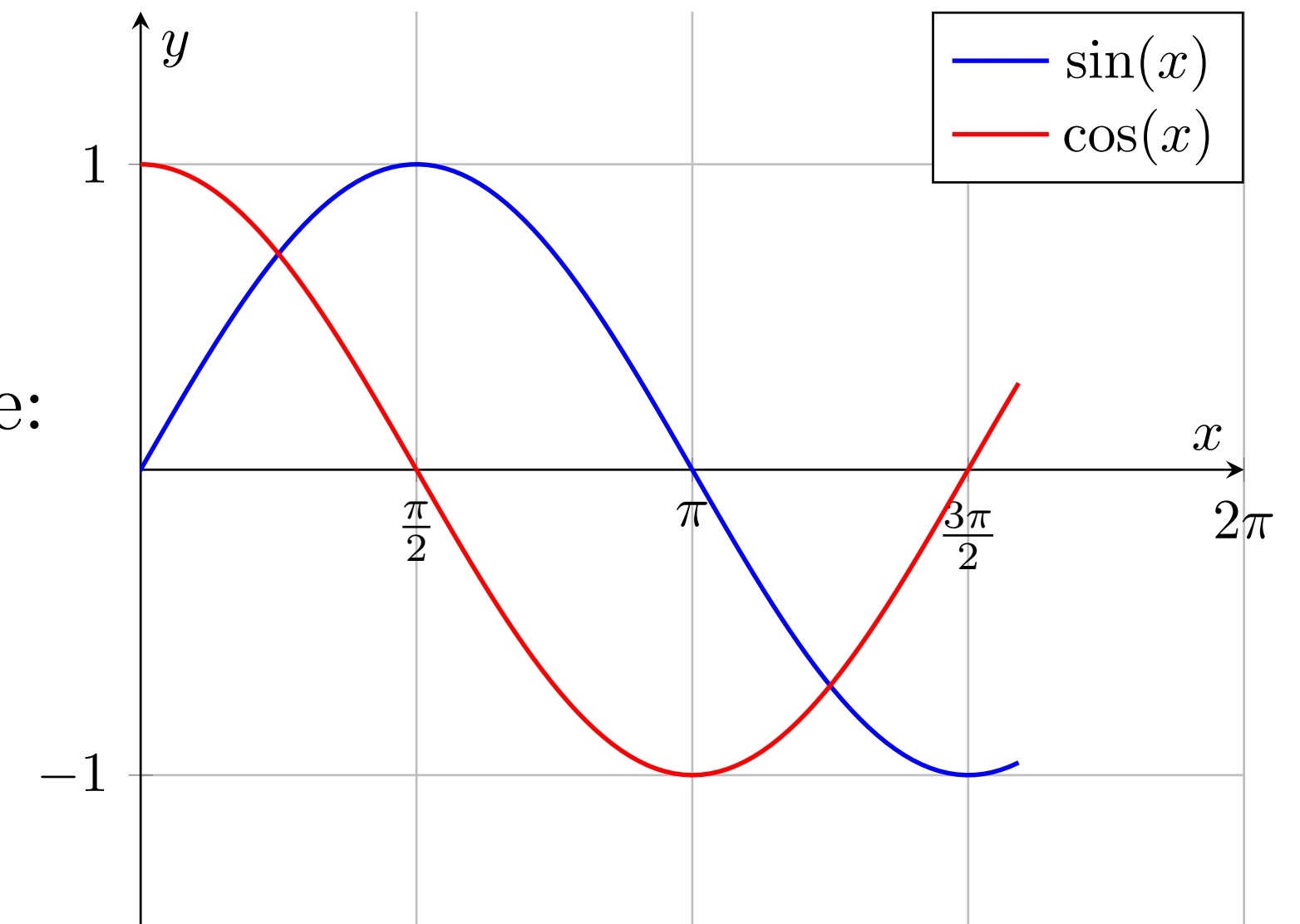
Evaluate each part:

$$\lim_{x \rightarrow \pi/2} 2x = \pi, \quad \lim_{x \rightarrow \pi/2} \sin \left(\frac{3\pi}{2} + x \right) = \sin(2\pi) = 0.$$

So the inside limit is $\pi + 0 = \pi$, hence

$$\lim_{x \rightarrow \pi/2} \cos \left(2x + \sin \left(\frac{3\pi}{2} + x \right) \right) = \cos(\pi) = -1.$$

Answer: -1 .



Continuous extension to a point

Sometimes a function is not defined at a point, but it can be naturally extended to become continuous there.

Example: The function

$$f(x) = \frac{\sqrt{x+1}-1}{x}, \quad x \neq 0,$$

is continuous everywhere on its domain except at $x = 0$.

Unlike $1/x$, it has a finite limit as $x \rightarrow 0$. Multiply numerator and denominator by the conjugate to simplify:

$$f(x) = \frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} = \frac{x}{x(\sqrt{x+1}+1)} = \frac{1}{\sqrt{x+1}+1}, \quad x \neq 0.$$

Hence,

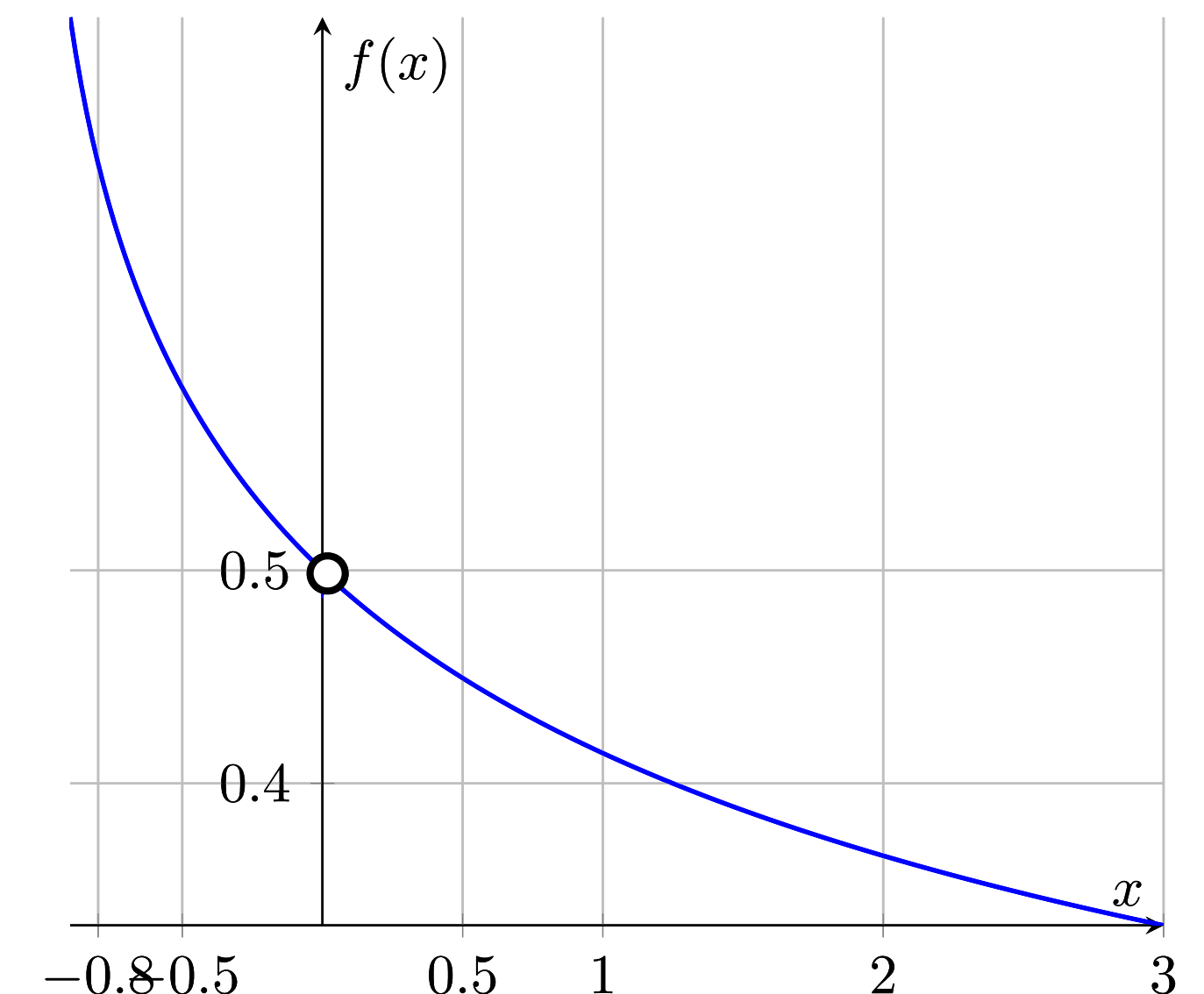
$$\lim_{x \rightarrow 0} f(x) = \frac{1}{1+1} = \frac{1}{2}.$$

We define the extended function

$$F(x) = \begin{cases} \frac{\sqrt{x+1}-1}{x}, & x \neq 0, \\ \frac{1}{2}, & x = 0, \end{cases}$$

which is continuous at $x = 0$ because

$$\lim_{x \rightarrow 0} F(x) = F(0) = \frac{1}{2}.$$



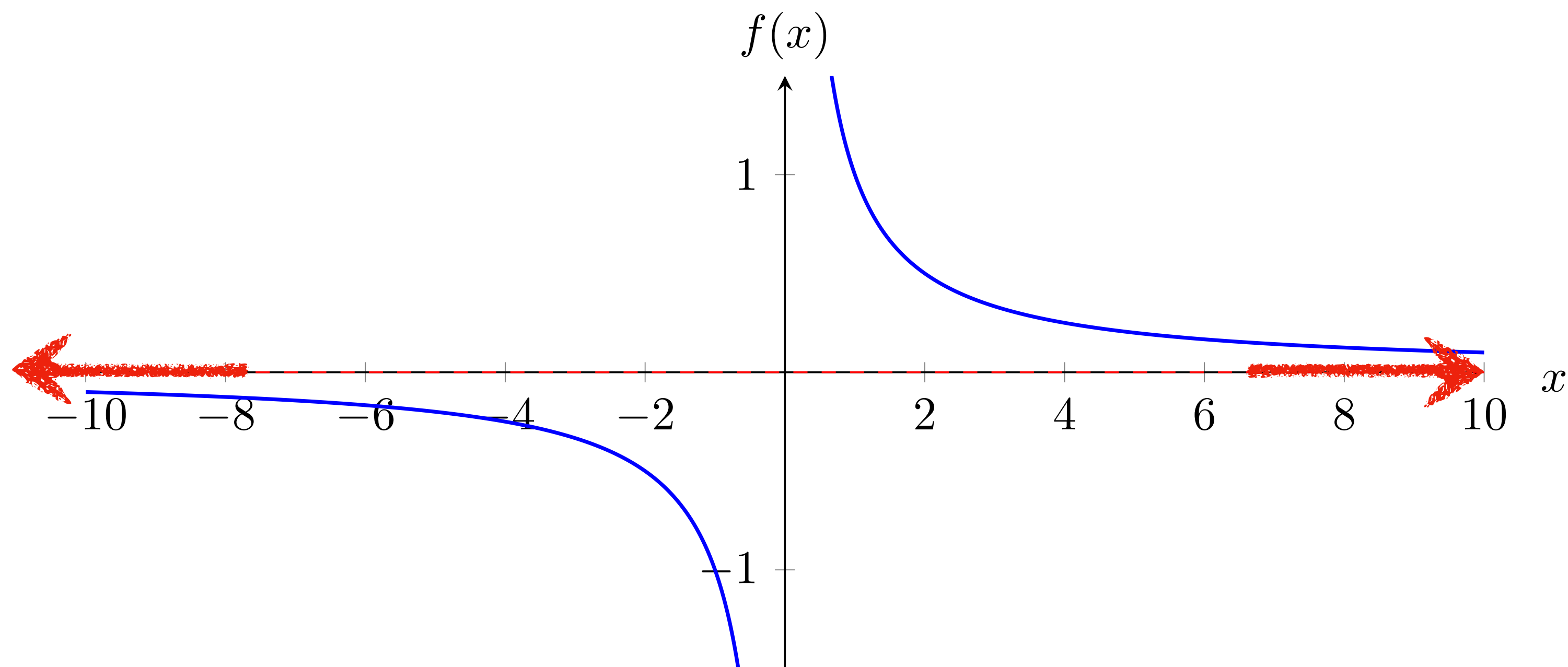
Limits at infinity

We often want to know how a function behaves as its input grows without bound, to understand long-term trends or horizontal asymptotes.

Example: Consider $f(x) = 1/x$. As $x \rightarrow \infty$ or $x \rightarrow -\infty$, $f(x) \rightarrow 0$:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

This motivates studying **limits at infinity**, which describe end-behavior of functions.



Limits at infinity

Definition: (Limits at Infinity)

Let $f(x)$ be a function defined for large $|x|$. Define $t = 1/x$. Then:

1. $f(x)$ has a limit L as $x \rightarrow \infty$ if the corresponding function $g(t) = f(1/t)$ has a **right-hand limit** L at $t = 0$:

$$\lim_{x \rightarrow \infty} f(x) = L \iff \lim_{t \rightarrow 0^+} g(t) = L.$$

2. $f(x)$ has a limit L as $x \rightarrow -\infty$ if the corresponding function $g(t) = f(1/t)$ has a **left-hand limit** L at $t = 0$:

$$\lim_{x \rightarrow -\infty} f(x) = L \iff \lim_{t \rightarrow 0^-} g(t) = L.$$

Example: Show that the following limits hold:

$$(a) \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad (b) \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Limits at infinity

Example: Show that the following limits hold:

$$(a) \lim_{x \rightarrow \infty} \frac{1}{x} = 0, \quad (b) \lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

Solution:

$$(a) \lim_{x \rightarrow \infty} \frac{1}{x}$$

Let $g(t) = f(1/t) = 1/(1/t) = t$. Then

$$\lim_{x \rightarrow \infty} \frac{1}{x} = \lim_{t \rightarrow 0^+} g(t) = \lim_{t \rightarrow 0^+} t = 0.$$

$$(b) \lim_{x \rightarrow -\infty} \frac{1}{x}$$

Again, let $g(t) = f(1/t) = t$. Now $x \rightarrow -\infty$ corresponds to $t \rightarrow 0^-$, so

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = \lim_{t \rightarrow 0^-} g(t) = \lim_{t \rightarrow 0^-} t = 0.$$

Thus, in both cases, the limit is 0.

Limits at infinity

Example: Find $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16})$.

Solution. Multiply and divide by the conjugate to simplify:

$$\begin{aligned} x - \sqrt{x^2 + 16} &= \frac{(x - \sqrt{x^2 + 16})(x + \sqrt{x^2 + 16})}{x + \sqrt{x^2 + 16}} = \frac{x^2 - (x^2 + 16)}{x + \sqrt{x^2 + 16}} \\ &= \frac{-16}{x + \sqrt{x^2 + 16}}. \end{aligned}$$

As $x \rightarrow \infty$, both x and $\sqrt{x^2 + 16}$ grow without bound, so the denominator $\rightarrow \infty$. Hence

$$\lim_{x \rightarrow \infty} (x - \sqrt{x^2 + 16}) = 0.$$

Infinite limits

Infinite limits capture when a function “blows up or down” without bound near a given point.

Limit equals $+\infty$:

- $\lim_{x \rightarrow c} f(x) = +\infty$
- As x gets close to c , $f(x)$ grows larger and larger.
- No matter how high you set a threshold, $f(x)$ eventually goes above it.

Limit equals $-\infty$:

- $\lim_{x \rightarrow c} f(x) = -\infty$.
- As x gets close to c , $f(x)$ decreases without bound.
- No matter how low you set a threshold, $f(x)$ eventually goes below it.

Definition: A line $x = a$ is a vertical asymptote of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty$$

Infinite limits

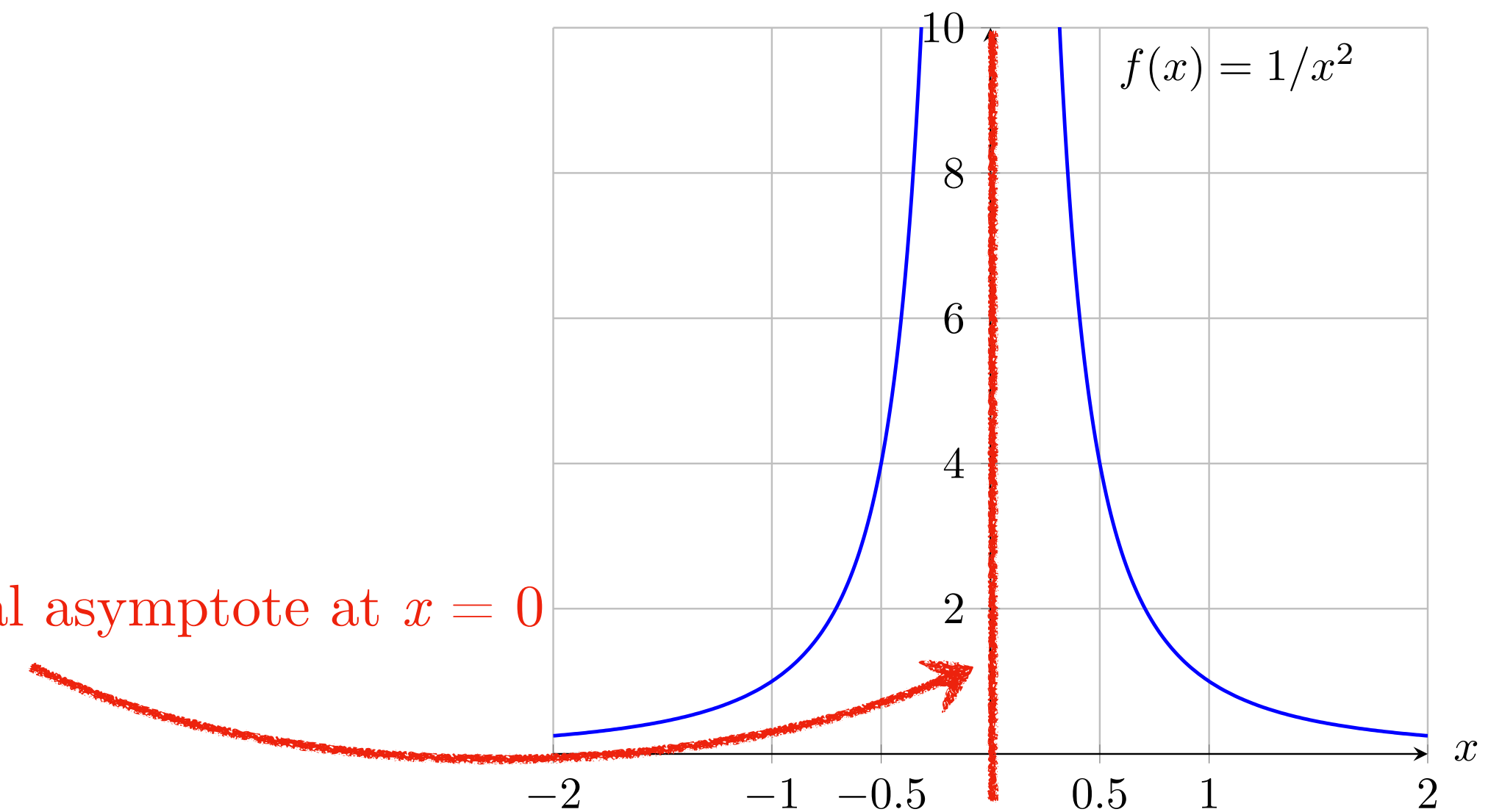
Example: Consider $f(x) = \frac{1}{x^2}$.

- As $x \rightarrow 0$, $f(x)$ grows without bound:

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

- The graph has a vertical asymptote at $x = 0$.

vertical asymptote at $x = 0$



Example: Consider $f(x) = \frac{1}{x-1}$

$$\lim_{x \rightarrow 1^+} \frac{1}{x-1} = +\infty, \quad \lim_{x \rightarrow 1^-} \frac{1}{x-1} = -\infty$$

- Shows different behavior when approaching $x = 1$ from the left or right.
- The graph has a vertical asymptote at $x = 1$.

vertical asymptote at $x = 1$

